

Declarations and Statements

Each section below must be included in the final manuscript before submission. Some go on the title page, some at the end before References — RSE-specific placement noted.

Declaration of Competing Interests

(Place: end of manuscript, before References.)

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding Declaration

(Place: end of manuscript, before Acknowledgements.)

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Update if any of the following apply: - PhD fellowship from ICAR / UGC / DST / state government - Project-specific grant - Institutional support from your department

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

(Place: a new section immediately before the References list, per Elsevier 2024 policy.)

During the preparation of this work the authors used Perplexity Computer (built on Anthropic's Claude Sonnet 4.6 large language model) in order to (i) conduct supplementary literature review, (ii) draft initial versions of the manuscript text and Google Earth Engine code, and (iii) generate figure templates and submission documentation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Per RSE policy, this declaration is mandatory if any AI tool was used beyond grammar and spell-checking. Honesty here is rewarded by reviewers; concealment, if discovered, is grounds for retraction.

Author Contributions (CRediT)

(Place: end of manuscript, before Funding.)

Role (CRediT)	Supranab Panda (ORCID 0009-0009-6496-6545)	Sarat Chandra Sahu (ORCID 0000-0002-8048-1910)
Conceptualization	Lead	Supporting
Methodology	Lead	Supporting
Software	Lead	—
Validation	Lead	Supporting
Formal analysis	Lead	Supporting
Investigation	Lead	—
Resources	Supporting	Lead
Data curation	Lead	—
Writing — original draft	Lead	—
Writing — review and editing	Lead	Supporting
Visualization	Lead	—
Supervision	—	Lead
Project administration	Lead	Supporting
Funding acquisition	—	—

CRediT statement (in publication-ready prose):

Supranab Panda: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review and editing, Visualization, Project administration. **Sarat Chandra Sahu:** Conceptualization, Methodology, Resources, Formal analysis, Validation, Writing — review and editing, Supervision.

Data Availability Statement

(Place: end of manuscript, after References, as a labeled section.)

All datasets used in this study are openly accessible without permission, application, or institutional gatekeeping. The Sentinel-1, Sentinel-2, JRC Global Surface Water, ERA5-Land, MODIS MCD12Q2, ESA WorldCover and GAUL datasets analysed in this study are openly available through Google Earth Engine. The IBTrACS cyclone track data are openly available from NOAA at <https://www.ncei.noaa.gov/products/international-best-track-archive>. ICRISAT Village Dynamics in South Asia (VDSA) microdata are openly available at <http://vdsa.icrisat.org>. District-level Kharif rice yield records are openly available from the Government of India Open Data Platform at <https://data.gov.in>. The RiceBaCI-GEE source code is released under the MIT licence at <https://github.com/pandasupranab/RiceBaCI-GEE>; permanent versioned snapshots (pre-registration, submission and acceptance releases) are archived at Zenodo via the GitHub–Zenodo integration at <https://doi.org/10.5281/zenodo.20024578>. The corrected 10-m SOS/POS/EOS rasters and validation reference points generated for this study are available on Mendeley Data at

<https://doi.org/10.17632/z3zxk4xy3c.1>. All Sentinel-2 reference imagery, coordinates, dates, labels, and label-confidence flags used for classifier validation are freely redistributable under the Copernicus Open Data licence and are included in full in the Mendeley Data deposit under CC-BY-4.0. A complete dataset manifest with download URLs and instructions is provided in docs/Data_Sources_Manifest.md.

Acknowledgements

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Submission Declarations

(Place: cover letter only, not in manuscript.)

By submitting this manuscript we confirm that:

- This work is original and has not been published previously.
- It is not currently under consideration for publication elsewhere.
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- All sources of funding and any potential conflicts of interest have been disclosed.